What is etcd?

Professional Answer

etcd is a distributed, highly available key-value store used to persist the entire cluster state, including configuration, metadata, and secrets.

Unprofessional Answer

etcd stores Kubernetes data.

What is etcd in Kubernetes architecture?

etcd is a distributed and consistent key-value store used to store the entire cluster state.
It acts as the single source of truth for configuration, metadata, and secrets.
etcd uses the Raft consensus algorithm to ensure strong consistency.
Only the kube-apiserver communicates directly with etcd.
If etcd data is lost, the Kubernetes cluster cannot be recovered.

Why is etcd critical to Kubernetes?

Professional Answer

etcd is the source of truth for the cluster; losing etcd means losing cluster state, configurations, and workloads.

Unprofessional Answer

Without etcd, Kubernetes won't work.

2. etcd (Cluster Memory Internals)

What it is internally

- Distributed key-value store
- Uses **RAFT consensus**
- Strong consistency (CP system)

Internal Working

- Stores everything as key-value pairs
- Example:
- /registry/pods/default/nginx

Key Internals

1. **Leader Election**
    - One leader
    - Followers replicate data
2. **Write Path**
    - Request → Leader → Majority acknowledgment → Commit
3. **Watch Mechanism**
    - API server and controllers watch etcd for changes
    - Enables event-driven architecture

Why etcd is critical

- Lose etcd → lose cluster state
- Backups are mandatory in production

What is etcd in Kubernetes?

Answer:
etcd is a **distributed, consistent, and highly available key-value store** used by Kubernetes to store **all cluster state**.
It acts as the **single source of truth** for Kubernetes.
Every object—Pods, Nodes, Secrets, ConfigMaps, Deployments—is stored in etcd.
If etcd is lost, the cluster loses its state.
Kubernetes control plane components continuously read from and write to etcd.

Why Kubernetes uses etcd?

Answer:
Kubernetes needs **strong consistency**, not eventual consistency.
etcd provides **linearizable reads and writes** using the **Raft consensus algorithm**.
It ensures all control plane components see the same cluster state.
etcd supports **watch functionality**, enabling real-time updates.
This makes scheduling, reconciliation, and self-healing possible.